Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] **[18EEOE03]**

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. List out various phases of Operation research and explain in brief any one of them. **07**

b. Discuss the assumptions made in Linear programing problem **07**

2. a. Write a brief note on "Transportation Problem". **04**

b. As the total demand is more than the total supply, it is not possible to meet the demand of every destination of the transportation problem given below: **10**

		Destination A	B	C	
Source	1	5	1	7	10
	2	6	4	6	80
	3	3	2	5	15
		75	20	50	

The penalty costs per unit of unsatisfied demand are Rs.5, 3 and 2 for destinations A, Band C respectively. Formulate the problem to minimize the total (transportation plus penalty) cost. Obtain basic feasible solution using different methods

3. a. Define Network, Event, Dummy activity, slack. **04**

b. Time estimates in weeks for PERT network is given below. Calculate the following: **10**

- **i.** Total expected time for the critical path
- **ii.** Standard deviation and variance for the project
- **iii.** Probability of project completion at least 4 weeks earlier than expected time
- **iv.** If the project due date is 19 weeks, what is the probability of not meeting the due date?

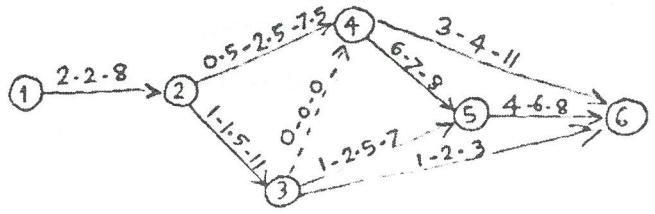
Activity	t_o	t_m	t_p
1 – 2	1	1	7
1 – 3	1	4	7
1 – 4	2	2	8
2 – 5	1	1	1
3 – 5	2	5	14
4 – 6	2	5	8
5 – 6	3	6	15

OR

4. a. State and explain Fulkerson's rule for numbering the event. **04**

b. For the network shown in Fig.below, the three time estimates for the activities are given along the arrows. Determine the critical path. What is the probability that the project will be completed in 20 days? **10**

Z	0.45	0.47	0.48	0.49	0.50
Ψ_Z	0.6736	0.6808	0.6844	0.6879	0.6915



5. a. Define Saddle point, Zero Sum game, Game Value. **04**

b. Reduce the game to either m x 2 or 2 x n by dominance, and then solve graphically. **10**

		B			
		B_1	B_2	B_3	B_4
A	A_1	19	6	7	5
	A_2	7	3	14	6
	A_3	12	8	18	4
	A_4	8	7	13	−1

OR

6. a. Explain maximin and minimax principle of Game theory. **04**

b. The following pay-off matrix is given for the two firms A and B. Use the dominance property to find the best mixed strategies for both the firms and find out the value of the game. **10**

Firm B

$$\text{Firm A} \quad \begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \end{array} \begin{array}{c} \begin{array}{ccc} B_1 & B_2 & B_3 \end{array} \\ \begin{bmatrix} 60 & 50 & 40 \\ 70 & 70 & 50 \\ 80 & 60 & 75 \end{bmatrix} \end{array}$$

7. a. Define forecasting. What are the steps involved in forecasting process? **04**

b. A company adopts method of least squares to develop a linear trend equation for the data as shown in the table below.

YEAR(X)	1	2	3	4	5	6	7	8	9	10	11
Shipment in tones (y)	2	3	6	10	8	7	12	14	14	18	19

Calculate the trend forecast for the year 12 and 20.

OR

8. **a.** List the various factors affecting forecasting. **04**

b. A firm use simple exponential smoothing with $\alpha = 0.1$ to forecast demand. The forecast for the week of February 1 was 500 units, whereas actual demand turned out to be 450 units.

i. Forecast the demand for the week of February 8 **10**

ii. Assume that the actual demand during the week of February 8 turned out to be 505 units. Forecast the demand for the week of February 15, Continue on Forecasting through March 15, assuming that subsequent demands were actually 516, 488, 467,554 and 510 units.